

Hamiltonian E.O.M: $\frac{d}{dt} \langle \hat{A} \rangle = -\frac{i}{\hbar} \langle \psi(t) | [\hat{A}, \hat{H}] | \psi(t) \rangle$

Operator form: $\frac{d}{dt} \hat{A}(t) = [\hat{A}, \hat{H}]$

$$\begin{aligned} \langle \psi(t) | \left(\frac{d}{dt} \hat{A}(t) \right) | \psi(t) \rangle &= \langle \psi(t) | [\hat{H}, \hat{A}] | \psi(t) \rangle \\ &= \frac{d}{dt} \langle \hat{A} \rangle - \left(\frac{d}{dt} \langle \psi(t) | \right) \hat{A} | \psi(t) \rangle - \langle \psi(t) | \hat{A} \left(\frac{d}{dt} | \psi(t) \rangle \right) \\ &= \frac{d}{dt} \langle \hat{A} \rangle - \left(\frac{d}{dt} \langle \psi(t) | \right) \hat{A} | \psi(t) \rangle - \\ &\quad \langle \psi(t) | \hat{A} \left(-\frac{i}{\hbar} \hat{H} | \psi(t) \rangle \right) \end{aligned}$$

As above, $\hat{A}$ is in the Schrödinger picture

so RHS = 0

$$\frac{d}{dt} \langle \hat{A} \rangle = -\frac{i}{\hbar} \langle \psi(t) | [\hat{A}, \hat{H}] | \psi(t) \rangle$$

• $|\psi\rangle = \sum_n a_n |\lambda_n\rangle \Rightarrow |\psi(t)\rangle = \sum_n a_n(t) |\lambda_n\rangle$

• $i\hbar \sum_n \dot{a}_n(t) |\lambda_n\rangle = \sum_n \hat{H} a_n(t) |\lambda_n\rangle$

$\leadsto$ Acting with $\langle \lambda_m |$

$$\sum_n i\hbar \dot{a}_n(t) \langle \lambda_m, \lambda_n \rangle = \sum_n a_n(t) \hat{H} |\lambda_n\rangle$$

$$i\hbar \sum_n \delta_{mn} \dot{a}_n(t) = \sum_n a_n(t) \hat{H} |\lambda_n\rangle$$

•• $\dot{a}_m(t) = (-i/\hbar) \sum_n a_n(t) \langle \lambda_m | \hat{H} | \lambda_n \rangle$

* $|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle$

$$\Rightarrow i\hbar \frac{d}{dt} [\hat{U}(t) |\psi(0)\rangle] = \hat{H} \hat{U}(t) |\psi(0)\rangle$$

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•• $\frac{d}{dt} \hat{U}(t) = -\frac{i}{\hbar} \hat{H} \hat{U}(t)$

* Has a formal solⁿ of the form

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar} \hat{H} t\right)$$

Subject to $\hat{H} \neq \hat{H}(t)$

Heisenberg Picture: $\hat{A}(t) = e^{-i\hat{H}t/\hbar} \hat{A} e^{i\hat{H}t/\hbar}$

$$\Rightarrow \frac{d}{dt} \hat{A}(t) = \frac{i}{\hbar} [\hat{H}, \hat{A}] + \frac{\partial}{\partial t} \hat{A}$$

Proof: $\frac{d}{dt} \langle \hat{A} \rangle = -\frac{i}{\hbar} \langle \psi(t) | [\hat{A}, \hat{H}] | \psi(t) \rangle$

$$\Rightarrow \text{RHS: } \frac{d}{dt} \langle \psi(0) | U^\dagger [\hat{A}, \hat{H}] U | \psi(0) \rangle$$

$$= i \hbar \langle \psi(0) | [\hat{A}(t), \hat{H}] | \psi(0) \rangle$$

$$\text{LHS: } \frac{d}{dt} \langle \psi(0) | \hat{A}(t) | \psi(0) \rangle$$

$$* \frac{d}{dt} (e^{-i\hat{H}t/\hbar} \hat{A} e^{i\hat{H}t/\hbar}) = -i\hat{H} e^{-i\hat{H}t/\hbar} \hat{A} e^{i\hat{H}t/\hbar} + U^\dagger \frac{\partial \hat{A}}{\partial t} U + e^{-i\hat{H}t/\hbar} \hat{A} i\hat{H} e^{i\hat{H}t/\hbar}$$

$$= -i\hat{H}\hat{A} + U^\dagger \left(\frac{\partial \hat{A}}{\partial t} \right) U + i\hat{A}\hat{H}$$

$$* = i[\hat{H}, \hat{A}] + U^\dagger \frac{\partial \hat{A}}{\partial t} U$$

$$\bullet \frac{d}{dt} \langle \hat{A} \rangle = \frac{d}{dt} \langle \psi(t) | \hat{A} | \psi(t) \rangle$$

$$= \frac{d}{dt} \langle \psi(0) | U^\dagger(t) \hat{A} U(t) | \psi(0) \rangle$$

$$= \langle \psi(0) | i\hat{H} U^\dagger \hat{A} U | \psi(0) \rangle$$

$$+ \langle \psi(0) | \frac{\partial \hat{A}}{\partial t} | \psi(0) \rangle$$

$$= \langle \psi(0) | U^\dagger \hat{A} i\hat{H} U | \psi(0) \rangle$$

$$= i \langle \psi(t) | [\hat{H}, \hat{A}] | \psi(t) \rangle + \langle \psi(t) | \frac{\partial \hat{A}}{\partial t} | \psi(t) \rangle$$

$$= \frac{d}{dt} \langle \hat{A} \rangle = i[\hat{H}, \hat{A}] + \frac{\partial \hat{A}}{\partial t}$$

Reduced density operator:

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i| \quad \leftarrow \text{Pure states}$$

$$\text{Tr}(\rho) = 1: \quad \text{Tr}(\rho) = \sum_l \langle l | \sum_i p_i |\psi_i\rangle \langle \psi_i| l \rangle$$

$$= \sum_i \sum_l p_i \langle \psi_i | l \rangle \langle l | \psi_i \rangle$$

$$= \sum_i p_i \langle \psi_i | \sum_l | l \rangle \langle l | \psi_i \rangle$$

$$= \sum_i p_i \langle \psi_i | \psi_i \rangle = \sum_i p_i = 1 \quad \checkmark$$

Partial Trace / Reduced trace: ρ_{AB}

$$\rho^A \equiv \text{Tr}_B(\rho^{AB})$$

Defined by:

$$\text{Tr}_B(|a_1\rangle \langle a_2| \otimes |b_1\rangle \langle b_2|) \equiv |a_1\rangle \langle a_2| \text{Tr}(|b_1\rangle \langle b_2|)$$

$$\text{Claim: } \text{Tr}(|b_1\rangle \langle b_2|) = \langle b_2 | b_1 \rangle$$

Proof: $\text{Tr}(|b\rangle\langle b|) = \sum_l \langle l|b\rangle\langle b|l\rangle$
 $= \langle b|(\sum_l |l\rangle\langle l|)|b\rangle$
 $= \langle b|b\rangle$ $\rho^A = \rho \mid \rho^B = \sigma$

ex: $\rho^{AB} = \rho \otimes \sigma$, then

$\rho^A = \text{Tr}_B(\rho \otimes \sigma) = \rho \text{tr}(\sigma) = \rho$

ex: Bell states: $(|00\rangle + |11\rangle)/\sqrt{2}$

$\rho = \left(\frac{|00\rangle + |11\rangle}{\sqrt{2}} \right) \left(\frac{\langle 00| + \langle 11|}{\sqrt{2}} \right)$
 $= \frac{1}{2} (|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$

$\rho^1 = \text{Tr}_2(\rho)$
 $= \text{Tr}_2(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$
 $= |0\rangle\langle 0| + |1\rangle\langle 1|$

$|00\rangle\langle 00| = |0\rangle\langle 0| \otimes |0\rangle\langle 0|$

* Somehow $|0\rangle\langle 1| = |1\rangle\langle 0|$

which leaves

$\rho^1 = (|0\rangle\langle 0| + |1\rangle\langle 1|)/2 = \frac{I}{2}$
 $|0\rangle\langle 0| + |1\rangle\langle 1| : \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

ex: $|0\rangle\langle 1| + |1\rangle\langle 0| : \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
 $|1\rangle\langle 0| : \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$

Proof of concept

$|1\rangle\langle 0| (a|0\rangle + b|1\rangle)$

$a|1\rangle$

$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ a \end{pmatrix} \Rightarrow$

$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$

$\begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$

$y_i^{\alpha\beta} = \frac{|i\rangle\langle i|}{\langle i|i\rangle} = \frac{|i\rangle\langle i|}{1} = |i\rangle\langle i|$

$\gamma_{ii}^{\alpha} = y_i^{\alpha\beta} / i = y_i^{\alpha\beta} \chi_i / i$

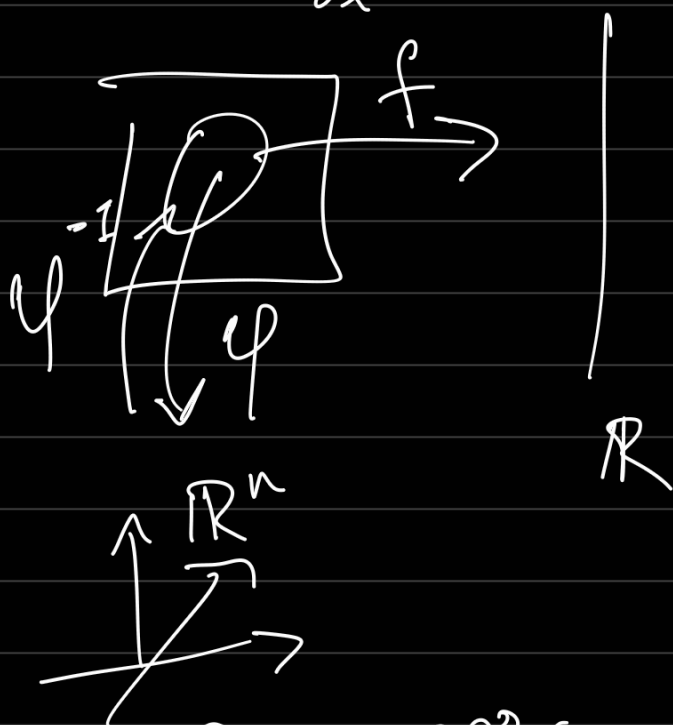
$$\langle i | \gamma_{i+1}^{\text{or}} = \left(\frac{|i+1\rangle \langle \mu_i|^\alpha - |i\rangle \langle \mu_{i+1}|^\alpha}{\langle i | i+1 \rangle} \right)$$

* Conformal field theory:

$$f: M \rightarrow \mathbb{R}$$

$$\bullet \quad f^* g = g \mid f^* g(x, y) = \sum^u \gamma^u g(f^* \partial_u, f^* \partial_x)$$

$$f_* (\partial_x)_p = \frac{\partial f(p)}{\partial x^u} \quad f(\phi) = \gamma$$



$$\partial_u = C^\infty(M) \rightarrow C^\infty(M)$$

$$\partial_u f = \frac{\partial x^u}{\partial x^u} \frac{\partial f}{\partial y^u}(P)$$